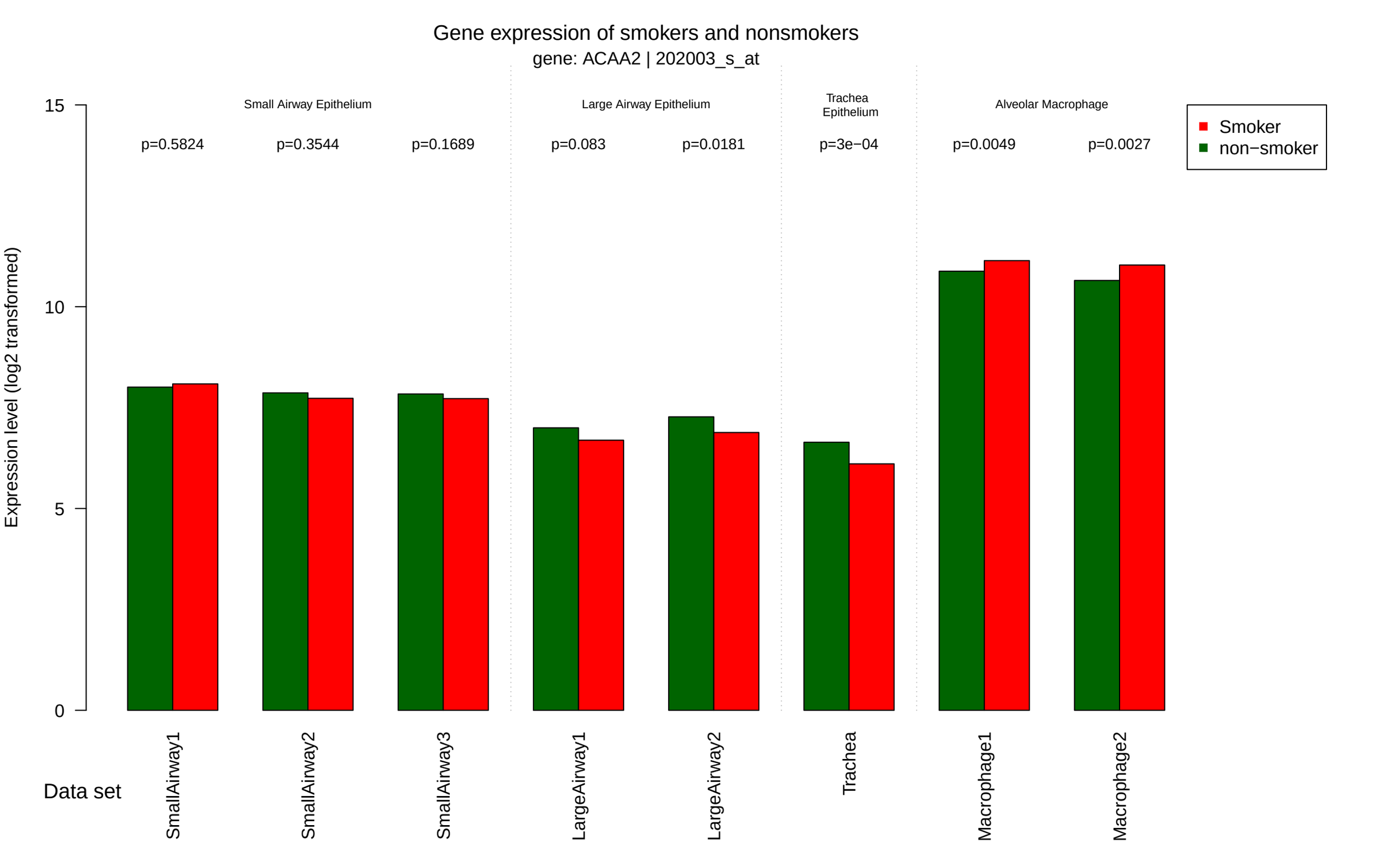
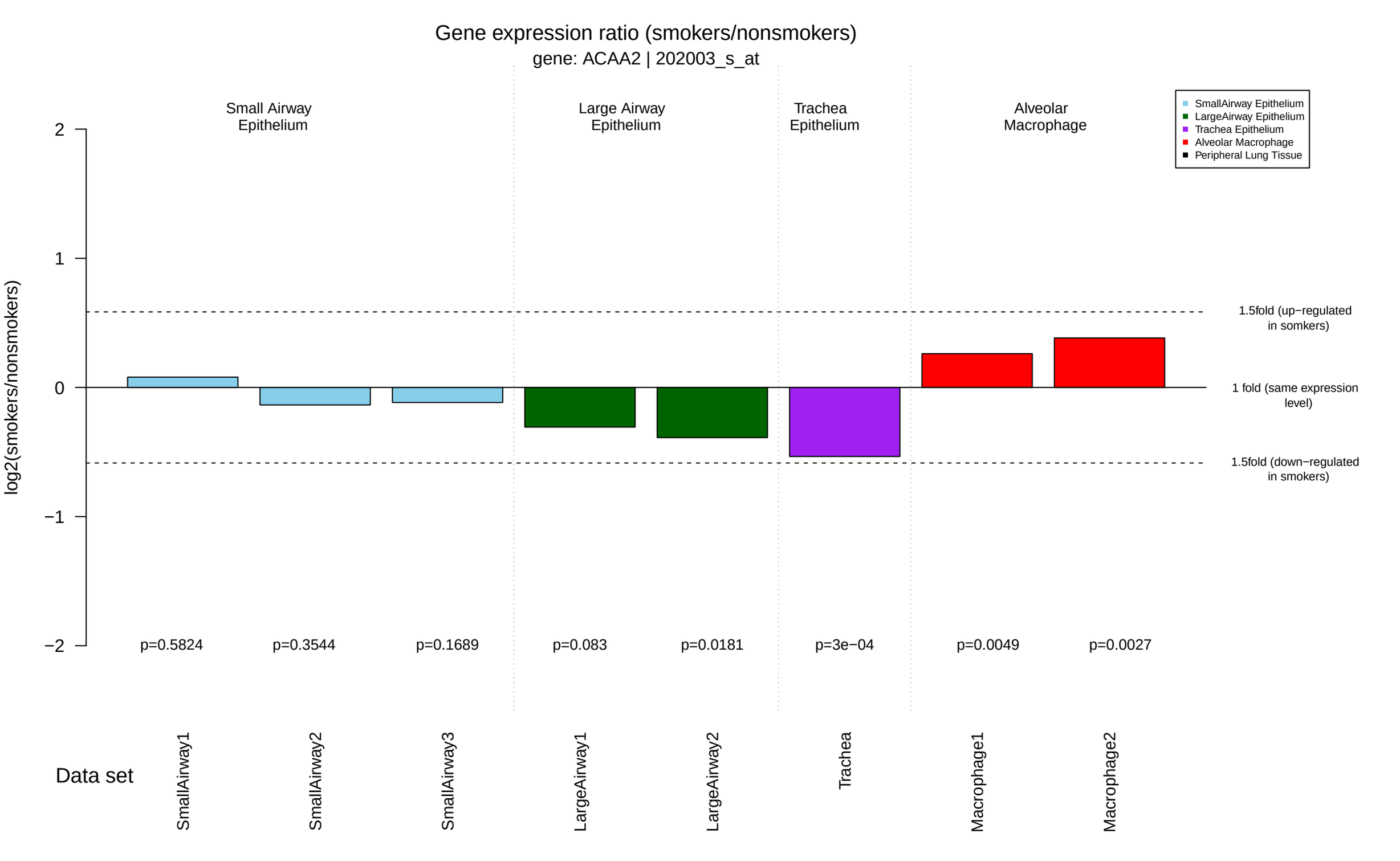
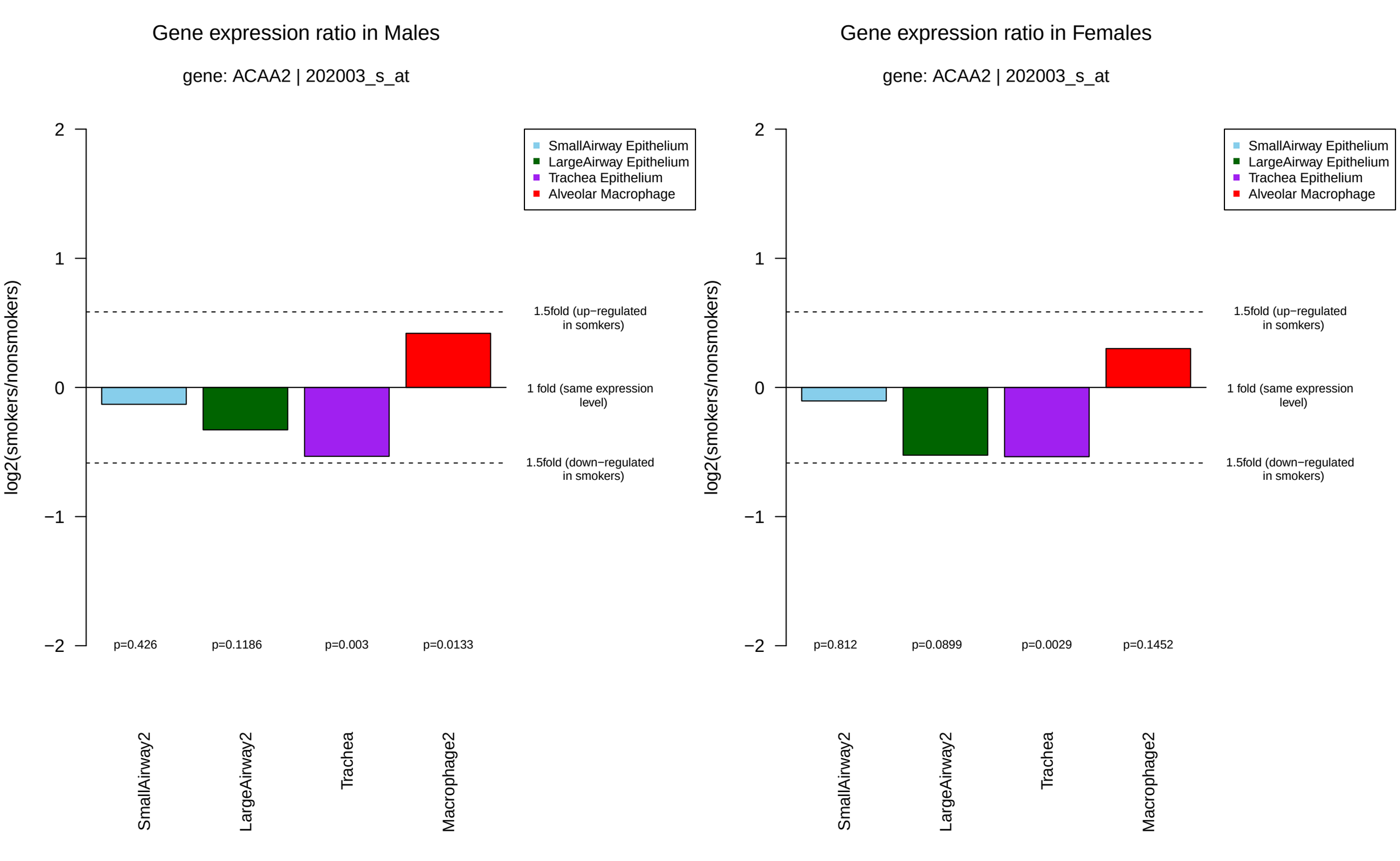
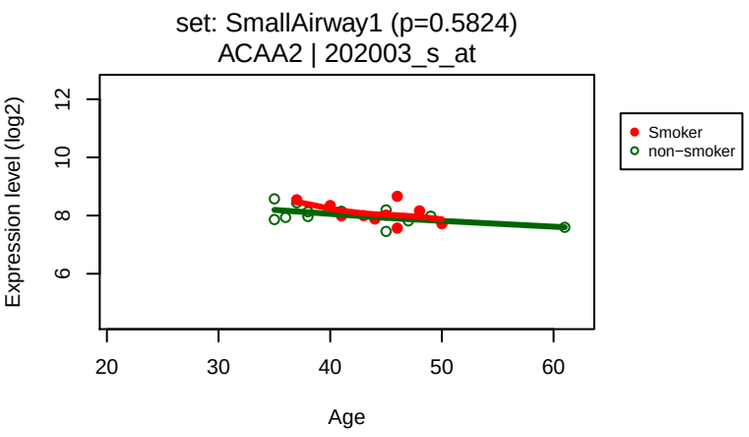
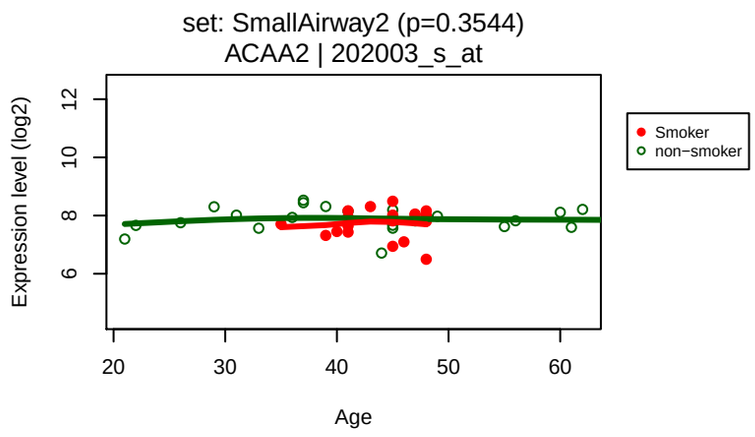
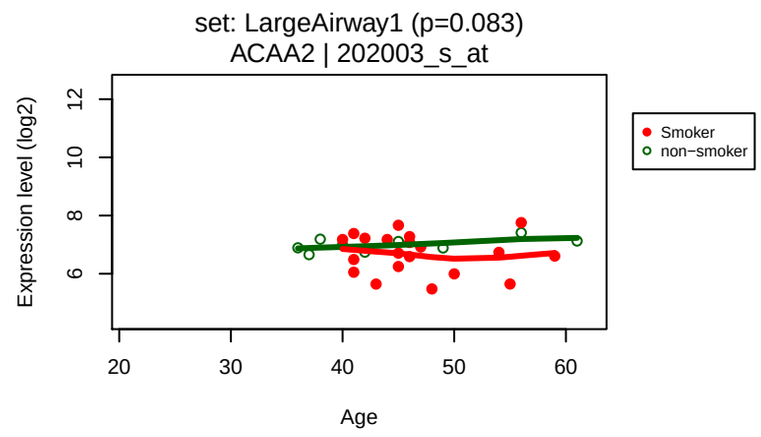

Gene expression of smokers and nonsmokers
gene: ACAA2 | 202003_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5824
p=0.3544
p=0.1689
p=0.083
p=0.0181
p=3e−04
p=0.0049
p=0.0027
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

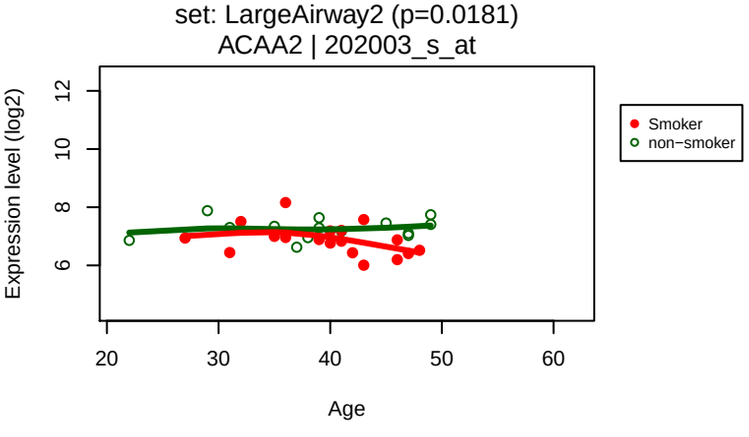

Gene expression ratio (smokers/nonsmokers)
gene: ACAA2 | 202003_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5824
p=0.3544
p=0.1689
p=0.083
p=0.0181
p=3e−04
p=0.0049
p=0.0027
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

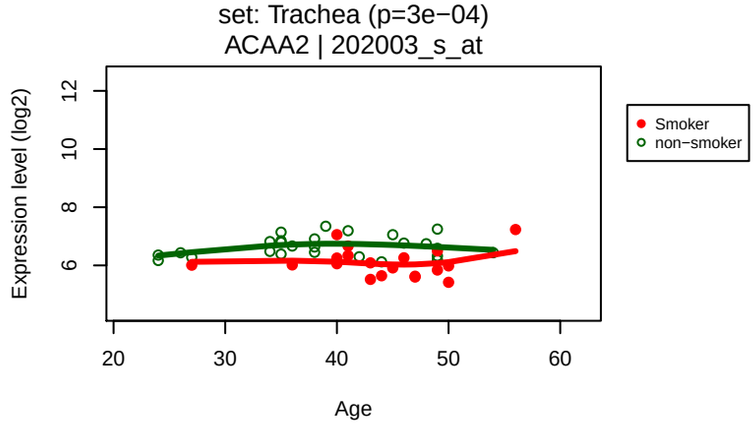

Gene expression ratio in Males
gene: ACAA2 | 202003_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.426
p=0.1186
p=0.003
p=0.0133
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAA2 | 202003_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.812
p=0.0899
p=0.0029
p=0.1452
SmallAirway2
LargeAirway2
Trachea
Macrophage2

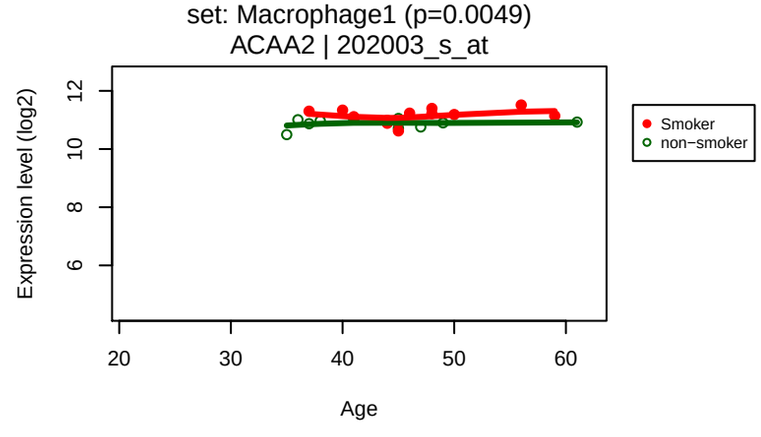

set: SmallAirway1 (p=0.5824)
ACAA2 | 202003_s_at
Expression level (log2)
Age
Smoker
non−smoker

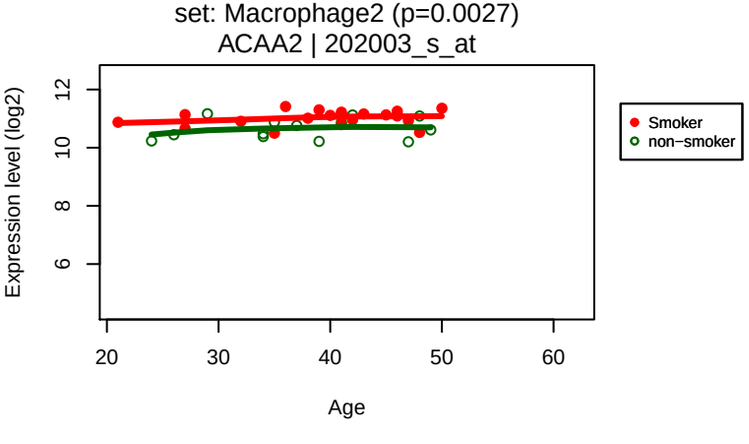

set: SmallAirway2 (p=0.3544)
ACAA2 | 202003_s_at
Expression level (log2)
Age
Smoker
non−smoker

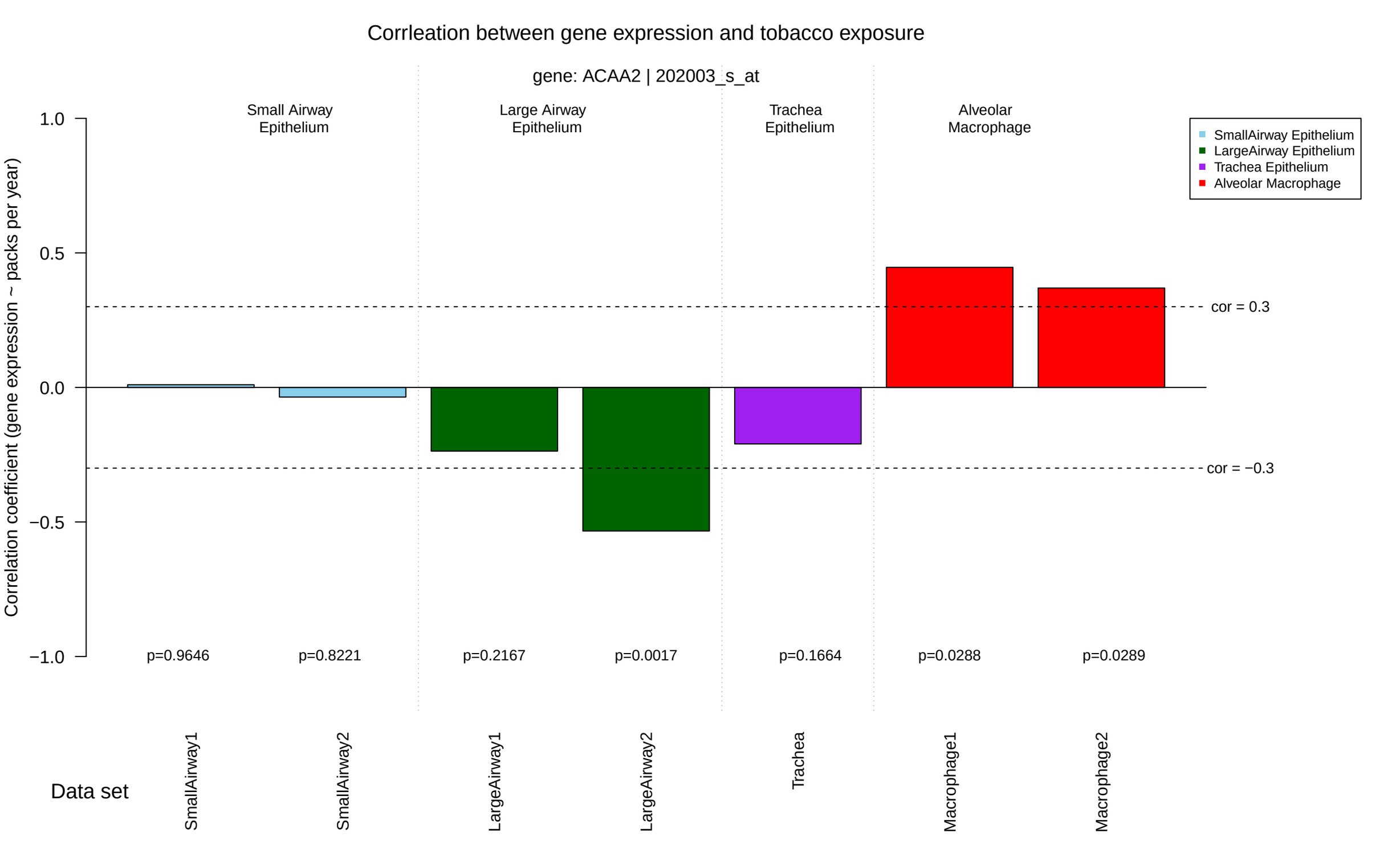

set: LargeAirway1 (p=0.083)
ACAA2 | 202003_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0181)
ACAA2 | 202003_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
ACAA2 | 202003_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0049)
ACAA2 | 202003_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0027)
ACAA2 | 202003_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAA2 | 202003_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9646
p=0.8221
p=0.2167
p=0.0017
p=0.1664
p=0.0288
p=0.0289
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2